

Hybrid Markov-Chain Monte Carlo - Controlled Random Walk Decoders in Scalable Qubit Architecture

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Question:

This proposal answers the question, "To what extent does a Hybrid Markov-Chain Monte Carlo + Controlled Random Walk (MCMC-CRW) Decoder increase fidelity through bias and local minima elimination in a large qubit depth architecture?"

Context:

Current quantum computing architecture uses around 1,000 qubits for universal application computers. In the future, as manufacturing efficiency and technology improve, the general trend is to increase the number of physical qubits per logical qubit for error correction purposes. Specifically, having more physical qubits increases the confidence in the error-corrected final state of a logical qubit. There are various methods for implementing error correction on these qubits, one of which is Monte Carlo decoding.

Monte Carlo decoding is based on the principle of discrete energy landscape sampling. It works by taking as many random samples as possible, evaluating them within the data set, and benchmarking optimization metrics such as their accuracy. Monte Carlo's searches across large data sets allow for a more accurate statistical prediction of an accurate decoder to select, searching the energy landscape for these regions. The accuracy of the convergence of this method of error configuration selection depends on the availability of data for it to sample. One issue associated with this is the tradeoff between computing time and accuracy. To increase the sample size and guarantee a higher accuracy, the software would face increased resource-intensiveness and latency to process and score each of the samples.

To minimize this issue of computing cost, the Markov-Chain Monte Carlo (MCMC) decoder was created, using a pre-made semi-optimal solution as a base, making minor changes to improve. The elimination of random shots exploring many possibilities leads to decreased computing time, less latency between results, and lower overhead costs. However, MCMC shares the same issue of local minima, along with an increased chance of bias from the selectiveness of samples used to run the algorithm.

Bias in an algorithm, such as error configuration selection, can manifest in many ways; one is the repeated selection of an imperfect configuration due to dependent sampling. Monte Carlo decoding and general statistics prefer the largest number of SRSs (Simple Random Sample) to get an accurate characterization of a population. However, MCMC doesn't take random samples in a traditional way, instead exploring smaller changes to achieve minimalistic optimization. This means that a sample for an MCMC decoder would show very small changes from the previous sample, meaning correlated sample-choice and exploration. Bias in MCMC arises from limited exploration due to poor mixing, when the Markov Chain becomes iteratively trapped in a subset of error configurations that are similar to each other, forming an endless loop without any improvement. Without enough mixing, the chain will not explore different changes, preventing it from escaping local minima.

To prevent this bias and allow an MCMC decoder to retain its minimal computing requirements, a method of randomness is proposed to be introduced into the system. Acting as a pseudo-classical Monte Carlo aimed at

increasing the exploration rate of the Markov chain, this addition creates a controlled random walk to break ties in ambiguous situations (a small difference between the best options for a selection). This would be very similar to teleporting a short distance away, across flat landscape, to check for other bigger holes. As the qubit depth of a circuit increases, the number of possible samples increases exponentially, and therefore, the number of ambiguous or “flat” landscapes would increase, providing more similar samples (flat ground). A traditional MCMC walk would frequently encounter bias and local minima, but a random ambiguity threshold between similarly-scored option allows for the exploration of options, and if properly tuned, would prevent bias and improve error configuration selection in future quantum architectures.

The importance of scalable decoders increases with the exploration of topological quantum computers, high-qubit-depth systems, and general advancements in quantum computing design and engineering. With stochastic decoders being a feasible method to implement scalable decoding, the importance of preventing bias in these decoders becomes prevalent; meaningfully using the improvements in QEC from larger qubit depth that would typically exacerbate the issues with stochastic decoding requires the improvement of traditional methods, creating the need for a hybrid MCMC - Controlled Random Walk framework.

Proposed Methodology:

To investigate the question, I propose that the relative comparison of various decoder types be used as a metric to determine the improvement derived from additions to the standard MCMC. Specifically, data for a large number of qubits, which can be changed over time based on computing constraints and the effectiveness of the decoder, could be randomly generated to simulate various error rates, along with their associated syndromes. Then, the data would be interpreted through 3 types of decoders: A union-find (and possibly an MWPM), a standard MCMC decoder, and the Hybrid MCMC-CRW. To evaluate the accuracy of these frameworks, the logical error rate could be used, and the runtime per computation or convergence step count could also be used as a weight or additional metric. Additionally, the logical error rate wouldn't present a large difference without significant trials, so I propose testing across various error frequencies to increase the chance of a logical fallacy occurring. To ensure that the true effect of a random walk is shown, the threshold for ambiguity selection must be tuned through experimentation.

To select different error configurations for the MCMC-CRW decoder, energy per sample could be used to select an optimal decoder. A general energy calculation function can be used to determine the energy per solution, based on negative log probabilities and Metropolis-Hastings selection. To implement the randomness, the energy of the previous proposal within a shot would be compared to the change in the next shot, and if sufficiently small, the proposed error after would be randomized, effectively creating another Markov Chain in a different location. When the energy margin is low enough to trigger the randomization algorithm, the lowest-energy proposal from the past before randomness is saved, and then compared across as many error explorations as the algorithm runs. The minimum energy per promising configuration would be compared, selecting the one with the lowest energy across the landscape.

Goal:

The goal of this proposal is to provide a more efficient alternative to a strong stochastic method for error decoding, allowing high efficiency without sacrificing excessive accuracy under time or computational constraints. This could be done using the IBM Qiskit SDK or any other simulation tool that adequately simulates the concepts.